

ASSIGNMENT-6

Case 1: Student Performance Report

A school wants an interactive report to analyze student performance by class, subject, and gender.

Assignment Tasks:

- Read student performance data provided in CSV file (student_performance_data.csv).
- Perform the visualizations on the given data:
 - Bar chart comparing subject averages.
 - Box plot of scores by gender.

```
[import matplotlib.pyplot as plt]
```

Case 2: Sales Performance Dashboard

You are a data analyst at an e-commerce company. The marketing head wants to track monthly sales trends, top-performing products, and revenue across regions.

Assignment Tasks:

- Import monthly sales data(ecommerce_sales_data.csv), having following attributes: e_commerce (Order_ID, Order_Date, Product_Name, Region, Quantity_Sold, Unit_Price, Revenue).
- Based on the given dataset create the following :
 - Line chart for monthly revenue trends.
 - Bar chart for top 10 best-selling products.
 - Pie chart for regional revenue distribution.
 - Heatmap of sales by category and month.
 - Create a scatter plot between Quantity_Sold and Revenue.

```
[import matplotlib.pyplot as plt]
```

Case 3: Customer Distribution Map for E-commerce Deliveries

You are a **Data Analyst at an e-commerce logistics company**. The operations team wants to visualize customer order locations across a city to identify **delivery hotspots**, **optimize delivery routes**, and **highlight regions with high order volume**.

Assignment:

Using customer order data with location coordinates generate a uniform random Latitude and Longitude coordinates, create an interactive map using **Folium** that:

1. Plots each order location as a marker or circle.
 2. Uses marker **color or size** to indicate order volume or revenue.
 3. Optionally adds clustering or heatmap to better visualize dense areas.
 - Includes a popup on each marker showing: Order ID, Revenue, Product or Region
- Use different colors for different regions.
 - Save the map as an HTML file for sharing.

[`import folium, import webbrowser, folium.Map, folium.Marker, folium.CircleMarker`]